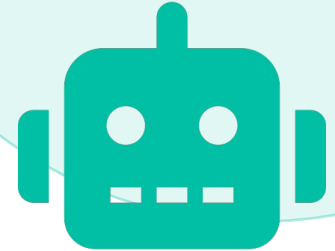


BrewBot

A Physical-State-Aware
Drink-Providing Robot

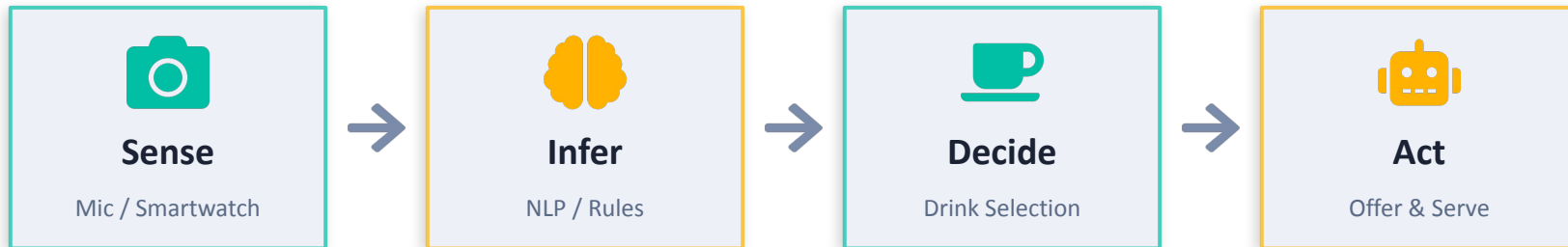
Yannick Martin · David Schmutz · Milos Veljovic



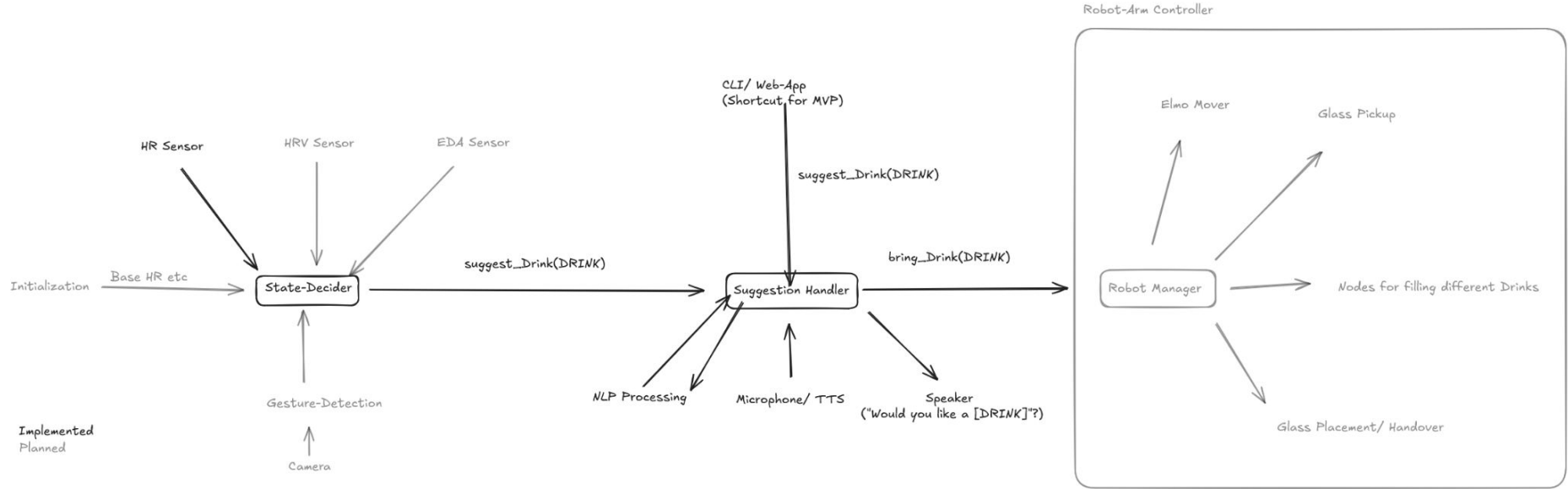
The BrewBot Idea



Motivation: Serving drinks in a lab is a low-stakes, naturalistic task. An autonomous robot that responds to physical state reduces participant burden and provides a concrete, evaluable HRI scenario.



System Pipeline



What Makes BrewBot Possible?



Multimodal Sensing

Microphones enable voice-command NLP and speech prosody analysis. Wearables (smartwatch/chest strap) provide continuous physiological signals — HR, HRV, and EDA.

NLP · Wearables



Physiological Signal Research

HR, HRV, and EDA are validated markers of stress, fatigue, and physical load. Rule-based thresholds enable reliable state classification without training data. [1,2]

HR · HRV · EDA



Trust Through Responsiveness

State-reactive robots achieve significantly higher user trust and engagement compared to static systems. [3]

HRI Research · 42 Papers

Which Sensor Detects Which State?



Microphone

Records speech audio: speech rate, volume & tonal features (prosody).
NLP extracts intent; rule-based stress detection via pitch and tempo.



Wearable

HR: heart rate in BPM
HRV: time between beats — RMSSD & LF/HF ratio.
EDA: skin conductance from sweat glands.



Camera

Tracks facial landmarks: skin tone & flushing (redness), blink rate & eye openness.
Body skeleton for posture analysis.
Frame-by-frame via MediaPipe.

Detectable States



Hot / Active

Very high EDA/ high HR

Cold Water



Stressed / Tense

Elevated HR / low HRV
under cognitive load

Tea



Fatigued / Tired

Low HRV + high EDA
(arousal marker)

Coffee



Cold

Low HR, low EDA (below
baseline)

Hot Chocolate



Needs Relaxing

After 16 o'clock

Beer

MVP & Potential Expansions

MVP



Robotic Basics

- Order a drink via web interface
- Only imitate filling glass
- Single drink option
- Elmo stays in place, only arm moves

V2



Human Interaction

- Rule-based state inference from physiological signals
- Robot proactively offers; participant confirms via NLP
- State→drink mapping: 2 options (e.g. water, coffee)
- Approach Sink/ Coffee Machine

V3



Cherry on top

- Camera for visual cues (facial flushing, posture, non-verbal communication, drop-off point)
- Real drinks?
- 3-4 Drink options
- Many small iterative improvements

Requirements



Hardware

HR Monitor

Smartwatch or chest strap.
Ideally includes HRV & EDA

Already covered:

Microphone, Speaker, Workstation (Laptop), Cameras



Additional Hardware (maybe)

EDA/GSR Sensor

If HR monitor lacks EDA

Local LLM

For speech-to-intent parsing beyond simple NLP.
Only if already available

(New) Uncertainties

Uncertainties

Sensor Calibration



HR/HRV/EDA baselines differ per person. A calibration period is needed. Current Plan: 5 minutes of calm sitting (resting HR), but may not be practical

Privacy



HR/HRV/EDA is medical data under GDPR. V3 feed to LLM adds sensitivity.

Open Questions



Coffee Machine Capabilities

Can the coffee machine provide drinks other than coffee?
E.g. hot water/ hot chocolate.



Ground Truth for Thresholds

How do we validate our HR/HRV/EDA rule thresholds? Self-report ratings during sessions can ground and calibrate rule-based classifications.



AprilTags on cylinders

Where to place them without problems?